# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 12 | Number of pieces = $n \div$ piece length. |
| 2 | **(c)** 15 | Scaling by a fraction is multiplication. |
| 3 | **(b)** 8 | Use the meaning of multiplying and dividing fractions. |
| 4 | **(d)** 6 | Convert to an improper fraction and multiply. |
| 5 | **(a)** $\frac{2}{3}$ | Write as an improper fraction, then invert. |
| 6 | **(d)** 2 | Multiply by the reciprocal of the divisor. |
| 7 | **(a)** $4\frac{1}{2}$ | Multiply the quantity by the number of batches. |
| 8 | **(d)** 14 | Number of pieces = $n \div$ piece length. |
| 9 | **(b)** 14 | Scaling by a fraction is multiplication. |
| 10 | **(c)** 2 | Use the meaning of multiplying and dividing fractions. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Count = $\frac{3}{4} \div \frac{1}{8}$; = $\frac{3}{4} \times \frac{8}{1} = \frac{24}{4}$ (1); Answer 6 (1). | 1+1 |
| 12 | $\frac{3}{4}$ of 48= 36; $\frac{2}{3} \times 36$ (1); = 24 kg (1). | 1+1 |
| 13 | $x \times \frac{2}{5} = \frac{4}{15}$; $x = \frac{4}{15} \div \frac{2}{5}$ (1); $x = \frac{2}{3}$ (1). | 1+1 |
| 14 | Count = $\frac{5}{2} \div \frac{5}{6}$; = $\frac{5}{2} \times \frac{6}{5} = \frac{30}{10}$ (1); Answer 3 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Area = $\frac{3}{4} \times \frac{2}{5} = \frac{3}{10}$ m$^2$ (1); Smaller side squared = $\frac{4}{25}$ m$^2$ (1); Difference = $\frac{7}{50}$ m$^2$ (1). | 1+1+1 |
| 16 | Improper form $\frac{5}{2}$; $\frac{5}{2} \times \frac{4}{3}$ (1); = $\frac{20}{6}$ (1); Answer $3\frac{1}{3}$ (1). | 1+1+1 |
| 17 | Cloth for 8 shirts = $8 \times \frac{5}{4}$; = 10 m (1); Shirts from 10\frac{5}{8} m = $8\frac{1}{2}$ (1); Full shirts = 8 (1). | 1+1+1 |
| | **OR:** Area = $\frac{5}{6} \times \frac{3}{10} = \frac{1}{4}$ m$^2$ (1); Smaller side squared = $\frac{9}{100}$ m$^2$ (1); Difference = $\frac{4}{25}$ m$^2$ (1). | 1+1+1 |
| 18 | Improper form $\frac{5}{3}$; $\frac{5}{3} \times \frac{6}{5}$ (1); = $\frac{30}{15}$ (1); Answer 2 (1). | 1+1+1 |
| | **OR:** Cloth for 6 shirts = $6 \times \frac{5}{2}$; = 15 m (1); Shirts from 16\frac{1}{4} m = $6\frac{1}{2}$ (1); Full shirts = 6 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | First part = $\frac{1}{4} \times 120 = 30$ km (1); Remaining = $120 - 30 = 90$ km (1); Second part = $\frac{2}{3} \times 90 = 60$ km (1); Left = $90 - 60$ (1); = 30 km (1). | 1+1+1+1+1 |
| | **OR:** Area = $\frac{4}{5} \times \frac{5}{6} = \frac{2}{3}$ (1); Vegetable part = $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ (1); Flower part = $\frac{2}{3} - \frac{1}{2}$ (1); = $\frac{1}{6}$ (1); State the three areas with units (1). | 1+1+1+1+1 |
| 20 | First part = $\frac{1}{3} \times 90 = 30$ km (1); Remaining = $90 - 30 = 60$ km (1); Second part = $\frac{3}{4} \times 60 = 45$ km (1); Left = $60 - 45$ (1); = 15 km (1). | 1+1+1+1+1 |
| | **OR:** Area = $\frac{3}{4} \times \frac{8}{9} = \frac{2}{3}$ (1); Vegetable part = $\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$ (1); Flower part = $\frac{2}{3} - \frac{4}{9}$ (1); = $\frac{2}{9}$ (1); State the three areas with units (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
- Inverting the wrong fraction when dividing: only the *divisor* is inverted, not the dividend.
- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change $\div$ to $\times$.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a $+$ sign.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 14 | Convert to an improper fraction and multiply. |
| 2 | **(b)** $\frac{4}{9}$ | Write as an improper fraction, then invert. |
| 3 | **(d)** 2 | Multiply by the reciprocal of the divisor. |
| 4 | **(b)** 6 | Multiply the quantity by the number of batches. |
| 5 | **(d)** 12 | Number of pieces $= n \div$ piece length. |
| 6 | **(c)** 15 | Scaling by a fraction is multiplication. |
| 7 | **(c)** $\frac{1}{2}$ | Use the meaning of multiplying and dividing fractions. |
| 8 | **(b)** 30 | Convert to an improper fraction and multiply. |
| 9 | **(c)** $\frac{3}{11}$ | Write as an improper fraction, then invert. |
| 10 | **(b)** 2 | Multiply by the reciprocal of the divisor. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\frac{5}{6}$ of 36$=30$; $\frac{3}{5} \times 30$ (1); $= 18$ litres (1). | 1+1 |
| 12 | $x \times \frac{3}{4} = \frac{9}{20}$; $x = \frac{9}{20} \div \frac{3}{4}$ (1); $x = \frac{3}{5}$ (1). | 1+1 |
| 13 | Count $= \frac{2}{3} \div \frac{1}{9}$; $= \frac{2}{3} \times \frac{9}{1} = \frac{18}{3}$ (1); Answer 6 (1). | 1+1 |
| 14 | $\frac{2}{7}$ of 70$=20$; $\frac{1}{2} \times 20$ (1); $= 10$ metres (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Area $= \frac{7}{8} \times \frac{4}{9} = \frac{7}{18}$ m$^2$ (1); Smaller side squared $= \frac{16}{81}$ m$^2$ (1); Difference $= \frac{31}{162}$ m$^2$ (1). | 1+1+1 |
| 16 | Improper form $\frac{10}{3}$; $\frac{10}{3} \times \frac{9}{5}$ (1); $= \frac{90}{15}$ (1); Answer 6 (1). | 1+1+1 |
| 17 | Cloth for 9 shirts $= 9 \times \frac{5}{3}$; $= 15$ m (1); Shirts from 15\frac{5}{6} m $= 9\frac{1}{2}$ (1); Full shirts $= 9$ (1). | 1+1+1 |
| | **OR:** Area $= \frac{2}{3} \times \frac{9}{10} = \frac{3}{5}$ m$^2$ (1); Smaller side squared $= \frac{4}{9}$ m$^2$ (1); Difference $= \frac{7}{45}$ m$^2$ (1). | 1+1+1 |
| 18 | Improper form $\frac{11}{4}$; $\frac{11}{4} \times \frac{15}{8}$ (1); $= \frac{165}{32}$ (1); Answer $5\frac{5}{32}$ (1). | 1+1+1 |
| | **OR:** Cloth for 12 shirts $= 12 \times \frac{9}{4}$; $= 27$ m (1); Shirts from 28\frac{1}{8} m $= 12\frac{1}{2}$ (1); Full shirts $= 12$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | First part $= \frac{3}{8} \times 160 = 60$ km (1); Remaining $= 160 - 60 = 100$ km (1); Second part $= \frac{2}{5} \times 100 = 40$ km (1); Left $= 100 - 40$ (1); $= 60$ km (1). | 1+1+1+1+1 |
| | **OR:** Area $= \frac{5}{6} \times \frac{6}{7} = \frac{5}{7}$ (1); Vegetable part $= \frac{4}{5} \times \frac{5}{7} = \frac{4}{7}$ (1); Flower part $= \frac{5}{7} - \frac{4}{7}$ (1); $= \frac{1}{7}$ (1); State the three areas with units (1). | 1+1+1+1+1 |
| 20 | First part $= \frac{2}{7} \times 84 = 24$ km (1); Remaining $= 84 - 24 = 60$ km (1); Second part $= \frac{3}{5} \times 60 = 36$ km (1); Left $= 60 - 36$ (1); $= 24$ km (1). | 1+1+1+1+1 |
| | **OR:** Area $= \frac{7}{8} \times \frac{4}{5} = \frac{7}{10}$ (1); Vegetable part $= \frac{5}{7} \times \frac{7}{10} = \frac{1}{2}$ (1); Flower part $= \frac{7}{10} - \frac{1}{2}$ (1); $= \frac{1}{5}$ (1); State the three areas with units (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
- Inverting the wrong fraction when dividing: only the *divisor* is inverted, not the dividend.
- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change $\div$ to $\times$.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a $+$ sign.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** $2\frac{1}{2}$ | Multiply the quantity by the number of batches. |
| 2 | **(a)** $8$ | Number of pieces $= n \div$ piece length. |
| 3 | **(d)** $21$ | Scaling by a fraction is multiplication. |
| 4 | **(a)** $4 \div \frac{1}{2}$ | Use the meaning of multiplying and dividing fractions. |
| 5 | **(c)** $7$ | Convert to an improper fraction and multiply. |
| 6 | **(b)** $\frac{5}{8}$ | Write as an improper fraction, then invert. |
| 7 | **(b)** $1\frac{1}{2}$ | Multiply by the reciprocal of the divisor. |
| 8 | **(d)** $2\frac{1}{2}$ | Multiply the quantity by the number of batches. |
| 9 | **(b)** $10$ | Number of pieces $= n \div$ piece length. |
| 10 | **(b)** greater than the number | Use the meaning of multiplying and dividing fractions. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $x \times \frac{5}{6} = \frac{25}{36}$; $x = \frac{25}{36} \div \frac{5}{6}$ (1); $x = \frac{5}{6}$ (1). | 1+1 |
| 12 | Count $= \frac{7}{4} \div \frac{7}{16}$; $= \frac{7}{4} \times \frac{16}{7} = \frac{112}{28}$ (1); Answer $4$ (1). | 1+1 |
| 13 | $\frac{8}{9}$ of $54 = 48$; $\frac{3}{4} \times 48$ (1); $= 36$ rupees (1). | 1+1 |
| 14 | $x \times \frac{7}{8} = \frac{21}{40}$; $x = \frac{21}{40} \div \frac{7}{8}$ (1); $x = \frac{3}{5}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Area $= \frac{5}{8} \times \frac{4}{15} = \frac{1}{6}$ m$^2$ (1); Smaller side squared $= \frac{16}{225}$ m$^2$ (1); Difference $= \frac{43}{450}$ m$^2$ (1). | 1+1+1 |
| 16 | Improper form $\frac{5}{4}$; $\frac{5}{4} \times \frac{12}{5}$ (1); $= \frac{60}{20}$ (1); Answer $3$ (1). | 1+1+1 |
| 17 | Cloth for 10 shirts $= 10 \times \frac{7}{4}$; $= 17\frac{1}{2}$ m (1); Shirts from 18\frac{3}{8} m $= 10\frac{1}{2}$ (1); Full shirts $= 10$ (1). | 1+1+1 |
|  | **OR:** Area $= \frac{3}{7} \times \frac{14}{15} = \frac{2}{5}$ m$^2$ (1); Smaller side squared $= \frac{9}{49}$ m$^2$ (1); Difference $= \frac{53}{245}$ m$^2$ (1). | 1+1+1 |
| 18 | Improper form $\frac{9}{2}$; $\frac{9}{2} \times \frac{8}{3}$ (1); $= \frac{72}{6}$ (1); Answer $12$ (1). | 1+1+1 |
|  | **OR:** Cloth for 9 shirts $= 9 \times \frac{10}{3}$; $= 30$ m (1); Shirts from 31\frac{2}{3} m $= 9\frac{1}{2}$ (1); Full shirts $= 9$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | First part $= \frac{2}{5} \times 150 = 60$ km (1); Remaining $= 150 - 60 = 90$ km (1); Second part $= \frac{5}{6} \times 90 = 75$ km (1); Left $= 90 - 75$ (1); $= 15$ km (1). | 1+1+1+1+1 |
|  | **OR:** Area $= \frac{9}{10} \times \frac{5}{6} = \frac{3}{4}$ (1); Vegetable part $= \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$ (1); Flower part $= \frac{3}{4} - \frac{3}{10}$ (1); $= \frac{9}{20}$ (1); State the three areas with units (1). | 1+1+1+1+1 |
| 20 | First part $= \frac{3}{10} \times 200 = 60$ km (1); Remaining $= 200 - 60 = 140$ km (1); Second part $= \frac{4}{7} \times 140 = 80$ km (1); Left $= 140 - 80$ (1); $= 60$ km (1). | 1+1+1+1+1 |
|  | **OR:** Area $= \frac{2}{3} \times \frac{3}{8} = \frac{1}{4}$ (1); Vegetable part $= \frac{5}{6} \times \frac{1}{4} = \frac{5}{24}$ (1); Flower part $= \frac{1}{4} - \frac{5}{24}$ (1); $= \frac{1}{24}$ (1); State the three areas with units (1). | 1+1+1+1+1 |

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- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
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